

ApexFlow Signal Trader Manual

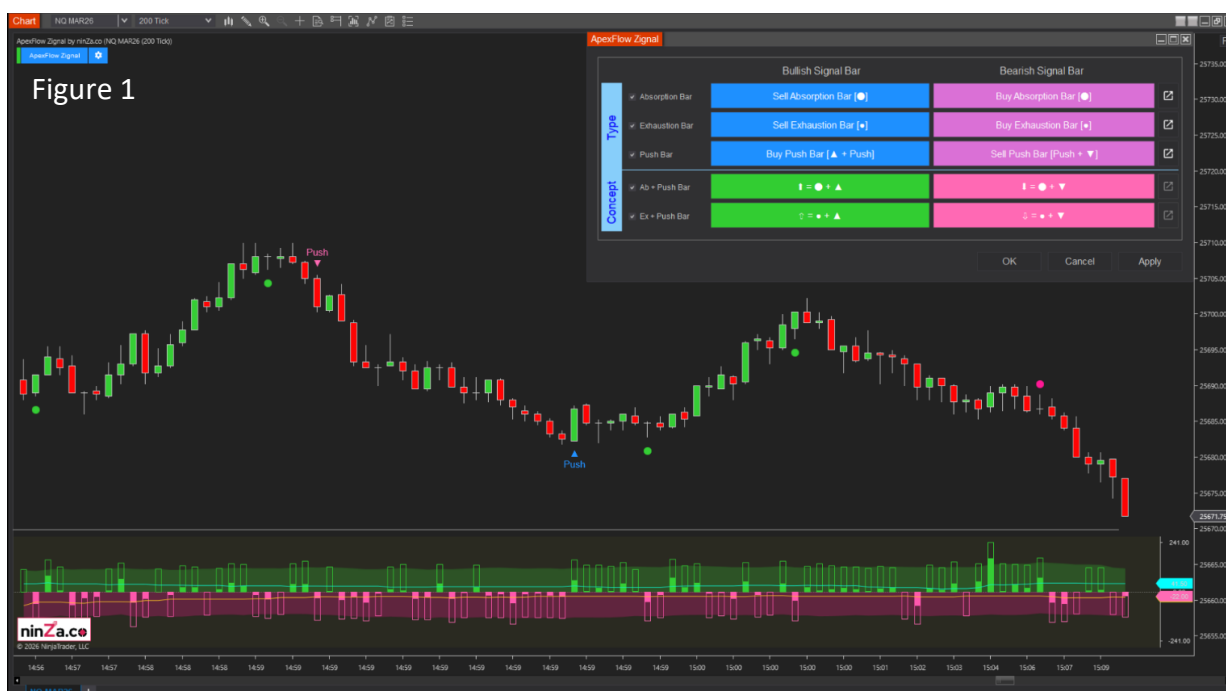
1. Overview

Dear trader,

Order Flow is one of the most powerful tools for understanding the market's actual behavior, yet it is also a complex data stream with large and constantly changing volumes of Buy/Sell/Delta information.

ApexFlow Signal is designed to transform raw Order Flow data into structured, behavior-based signals that are easier to see, easier to customize, and easier to apply in real trading conditions.

The system is built with No Tick-Replay technology, ensuring stable performance and a smooth trading experience even during fast-moving market conditions. You do not need to enable Tick Replay to use this indicator.



2. Signal mechanism

The signal mechanism of ApexFlow Zignal is built on **the Effort–Result principle** to interpret the actual behavior of order flow within each price bar.

Effort evaluates the level of buying or selling pressure, measured through:

- Volume
- Delta
- The POC and its position relative to the Neutral Range
- OHLC structure

Result reflects how price responds to the prior effort, evaluated through:

- OHLC structure
- Body and Full Range in relation to the context of N bars
- The closing position relative to the Neutral Range
- Volume

Based on the relationship between Effort and Result, ApexFlow Zignal classifies Order Flow behavior into distinct flow states and organizes them into 2 groups of signals:

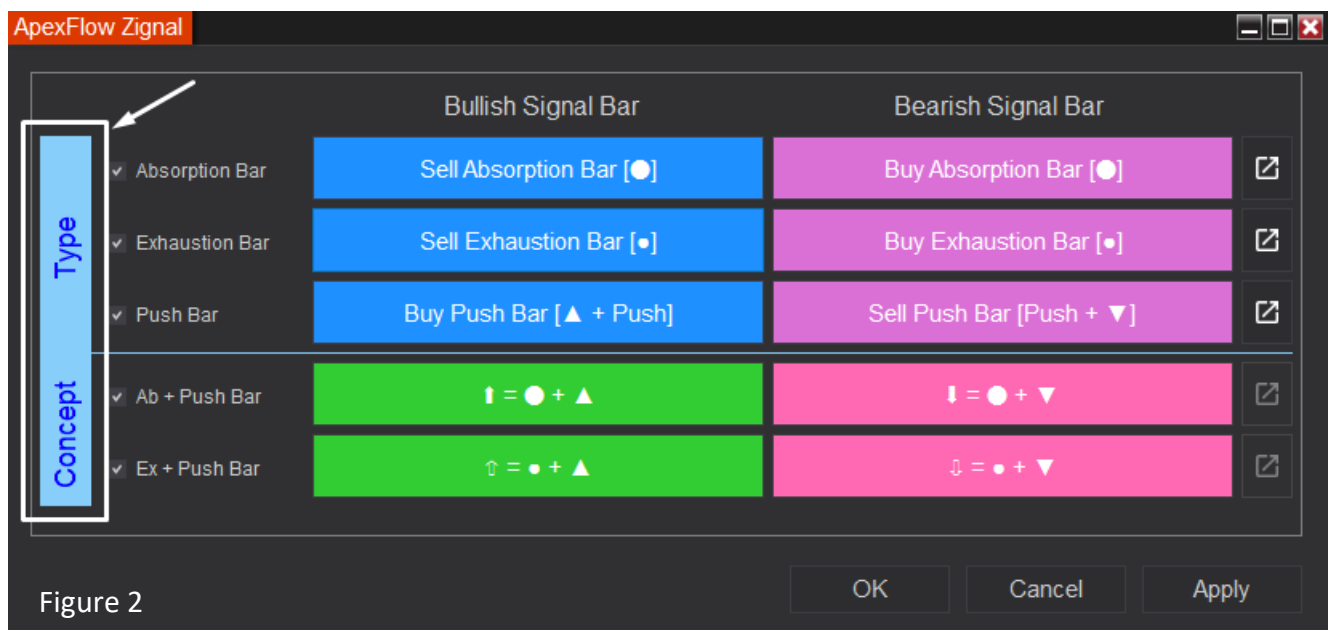


Figure 2

- **Signal Types**

These describe the Order Flow state within a single price bar, where the relationship between Effort and Result reveals either a clear imbalance or a clear alignment between buying and selling pressure.

- **Signal Concepts**

These combine multiple Signal Types in chronological sequence. By confirming the signal through the market's subsequent reaction, Signal Concepts increase reliability and help filter out noise.

2.1. Signal Types

ApexFlow Signal allows users to define Effort and Result by **enabling/disabling specific condition groups** within the signal configuration panel.

When you enable or disable a condition, the signal will appear based on – or will ignore – that condition accordingly.

N = 3

N: This parameter defines the lookback period (in number of bars) used to evaluate conditions relative to the surrounding bars.

This parameter will be explained in more detail in section 3.5 – Signal Period (N).

2.1.1. Absorption Bar

Figure 3

Absorption Bar Conditions

ApexFlow Signal

Concept	Type	Bullish Signal Bar	Bearish Signal Bar
✓ Absorption Bar		Sell Absorption Bar [●]	Buy Absorption Bar [●]
✓ Exhaustion Bar		Sell Exhaustion Bar [●]	Buy Exhaustion Bar [●]
✓ Push Bar		Buy Push Bar [▲ + Push]	Sell Push Bar [Push + ▼]
✓ Ab + Push Bar		▲ = ● + ▲	▲ = ● + ▼
✓ Ex + Push Bar		▼ = ● + ▲	▼ = ● + ▼

OK Cancel Apply

Arsorption Condition

Sell Absorption Bar [●]	Buy Absorption Bar [●]
Strong Sell effort, but the outcome favors the Buy side and is insignificant.	Strong Buy effort, but the outcome favors the Sell side and is insignificant.
N = 3	
Strong Sell effort	Strong Buy effort
✓ Negative Delta ✓ Negative Delta is above average threshold ✓ Sell Volume above average threshold ✓ Maximum Sell volume over N bars	✓ Positive Delta ✓ Positive Delta above average threshold ✓ Buy Volume above average threshold ✓ Maximum Buy volume over N bars
Insignificant Buy-side outcome	Insignificant Sell-side outcome
✓ Bullish bar ✓ Close price above neutral range ✓ Body size is not maximum over N bars ✓ Range is not maximum over N bars	✓ Bearish bar ✓ Close price below neutral range ✓ Body size is not maximum over N bars ✓ Range is not maximum over N bars

OK Cancel Apply

Effort conditions

Outcome conditions

An **Absorption Bar** represents a bar where the Effort is high but the Result is disproportionately small.

In an Absorption Bar, one side of the market (Buy or Sell) shows a **clear trading effort**, yet price does not respond in the corresponding direction.

The price movement is restricted, indicating that this effort has been **absorbed**, and the eventual outcome favors the **opposite side**—but only marginally

- **Sell Absorption Bar** appears when Sell pressure is dominant, but price does not decline accordingly, and the minor result favors the Buy side.

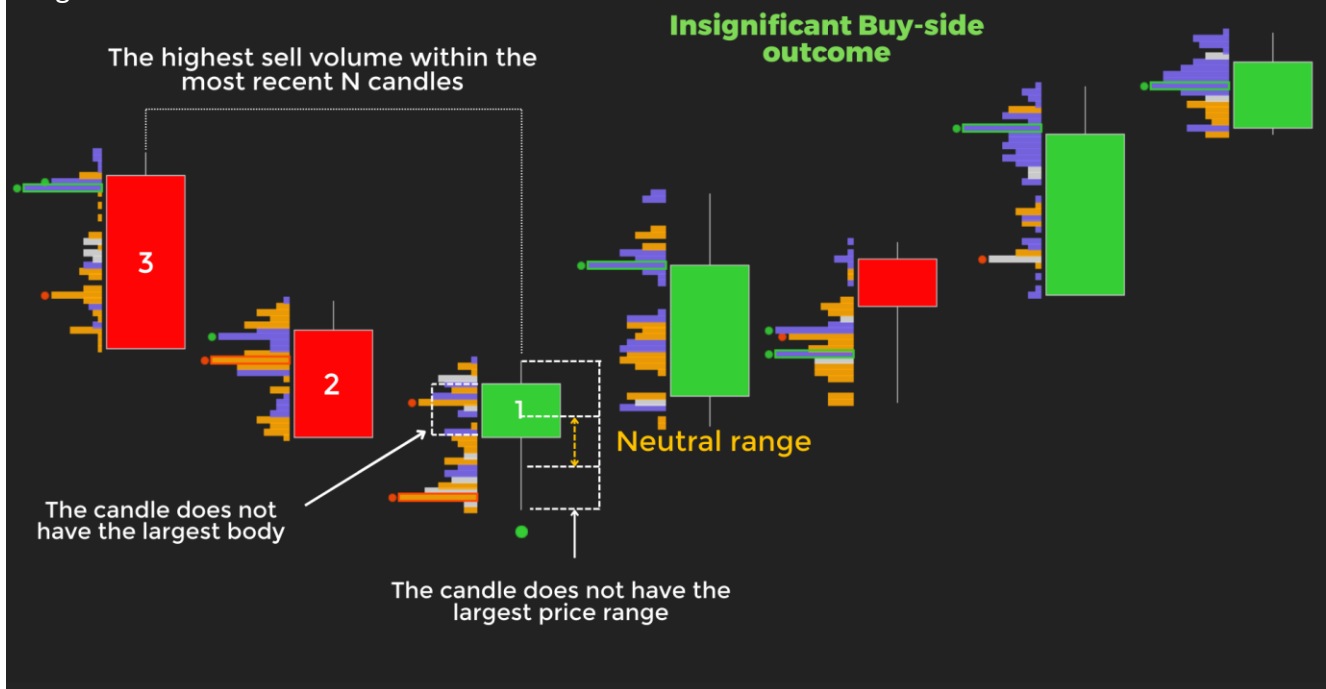
A Sell Absorption Bar is a **bullish** signal.



Strong Sell Effort is reflected through conditions such as:

- Negative delta
- Negative delta exceeding its average threshold
- Sell volume exceeding its average threshold
- Highest Sell volume within the recent N-bar context

Figure 5



The **insignificant Buy-side outcome** is reflected through the following conditions:

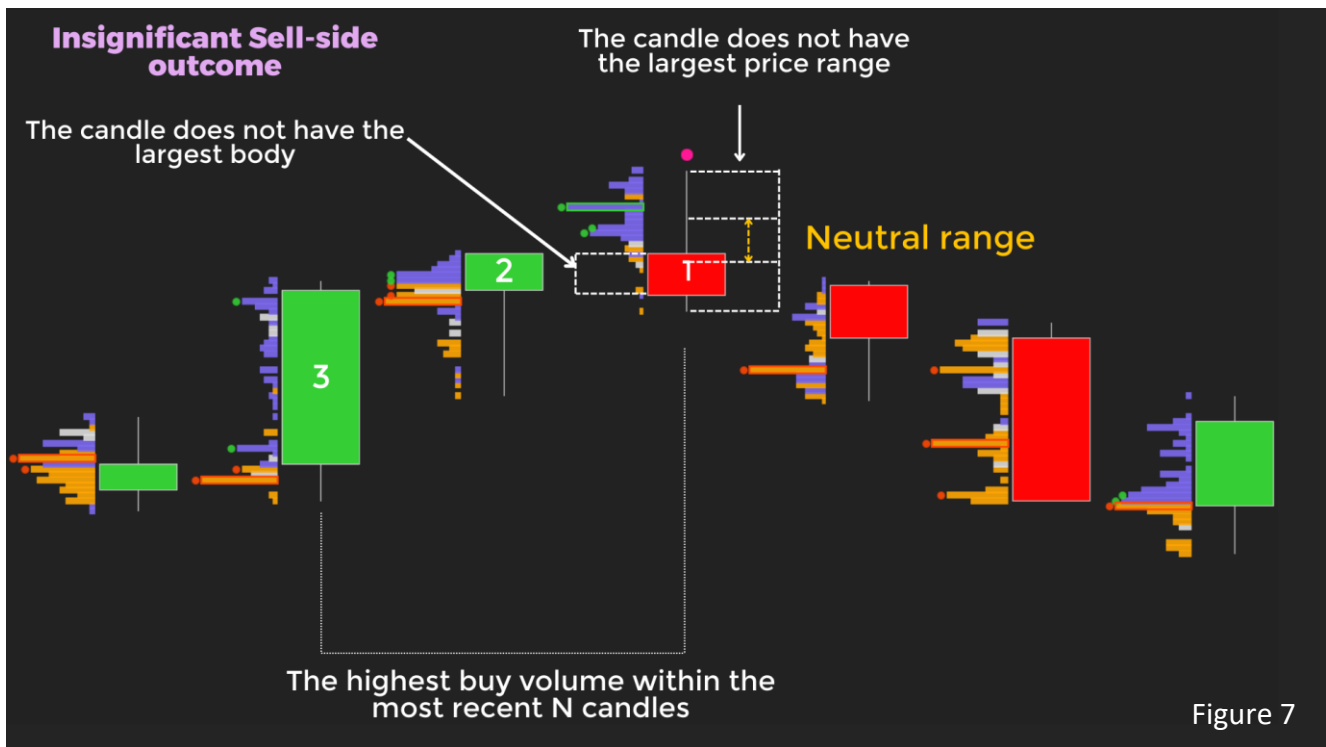
- The bar is bullish
- The closing price is above the Neutral Range
- The bar does **not** have the largest body within the recent N-bar context
- The bar does **not** have the largest full range within the recent N-bar context
- **Buy Absorption Bar** appears when Buy pressure is dominant, but price does not rise accordingly, and the minor outcome favors the Sell side.

A Buy Absorption Bar is a **bearish** signal.



Strong Buy Effort is reflected through conditions such as:

- Positive delta
- Positive delta exceeding its average threshold
- Buy volume exceeding its average threshold
- Highest Buy volume within the recent N-bar context



The **insignificant Sell-side outcome** is reflected through the following conditions:

- The bar is bearish
- The closing price is below the Neutral Range
- The bar does **not** have the largest body compared with the recent N bars
- The bar does **not** have the largest full range compared with the recent N bars

2.1.2. Exhaustion Bar

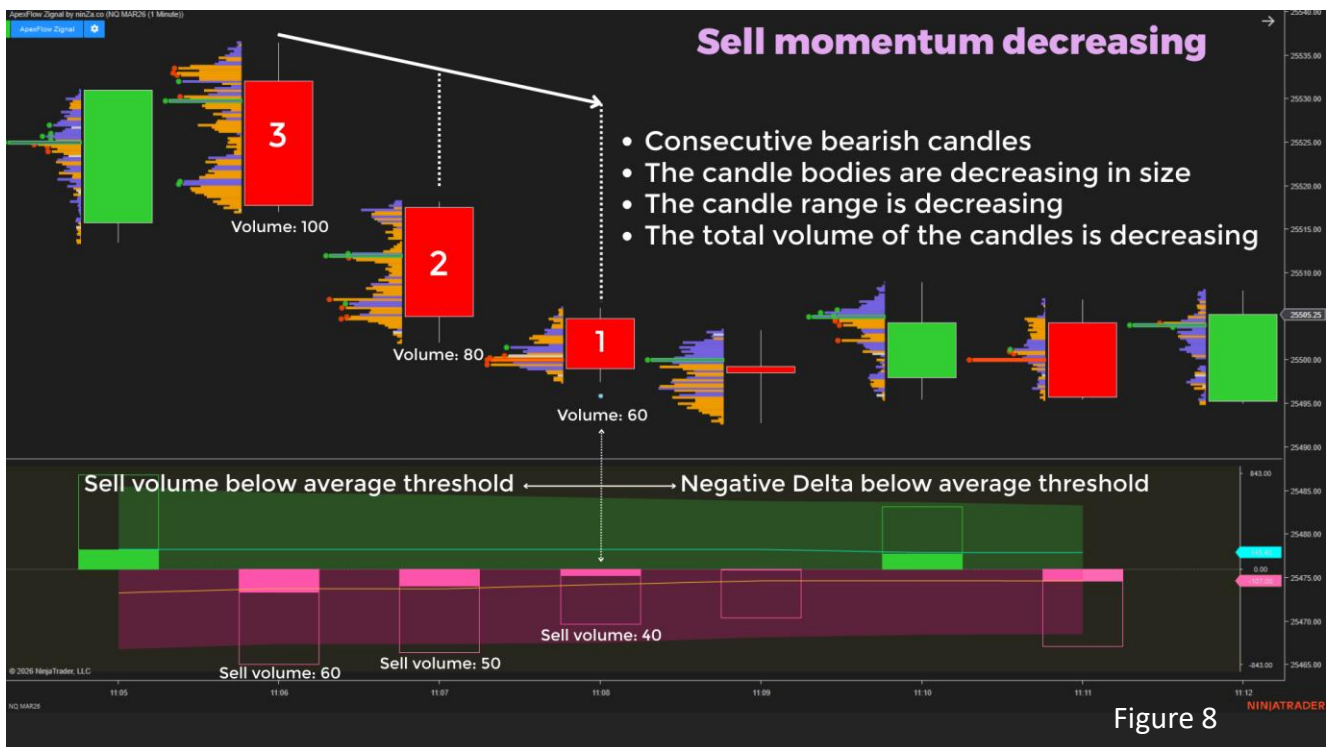
An **Exhaustion Bar** represents a state where the dominant side still shows active effort, but momentum is clearly weakening over time.

In an Exhaustion Bar, the Effort can no longer maintain the quality of the Result, indicating that the market is losing momentum and may be preparing to shift its state.

- **Sell Exhaustion Bar** appears when selling pressure is still present but sell-side momentum is weakening and no longer strong enough to push price downward

effectively.

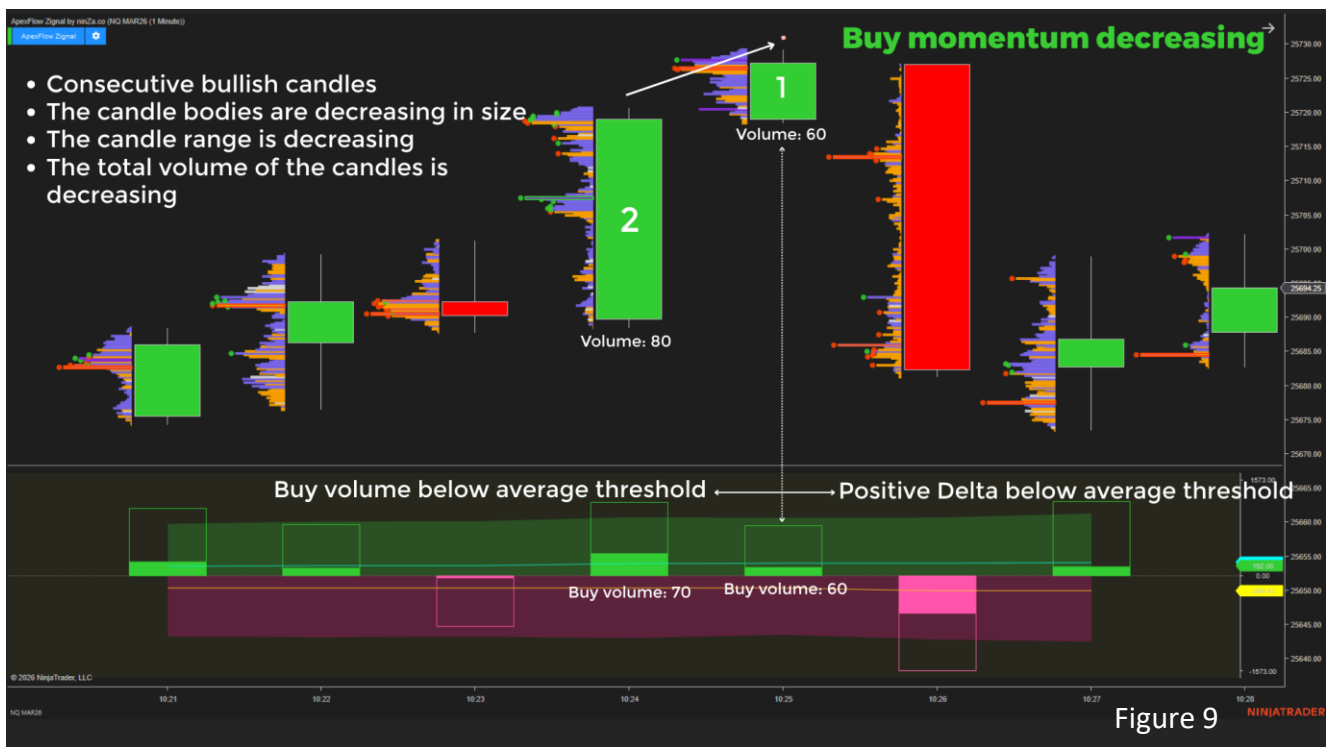
A Sell Exhaustion Bar is a **bullish** signal.



Decreasing Sell-side momentum is reflected through conditions such as:

- A sequence of N consecutive bearish bars
- A sequence of N bars with progressively smaller bearish bodies
- A sequence of N bars with progressively smaller full ranges
- Total volume decreasing within the recent N-bar context
- Sell volume decreasing within the recent N-bar context
- Negative delta exceeding its average threshold
- Sell volume exceeding its average threshold
- **Buy Exhaustion Bar** appears when buying pressure is still present, but Buy-side momentum is weakening and no longer strong enough to push price upward effectively.

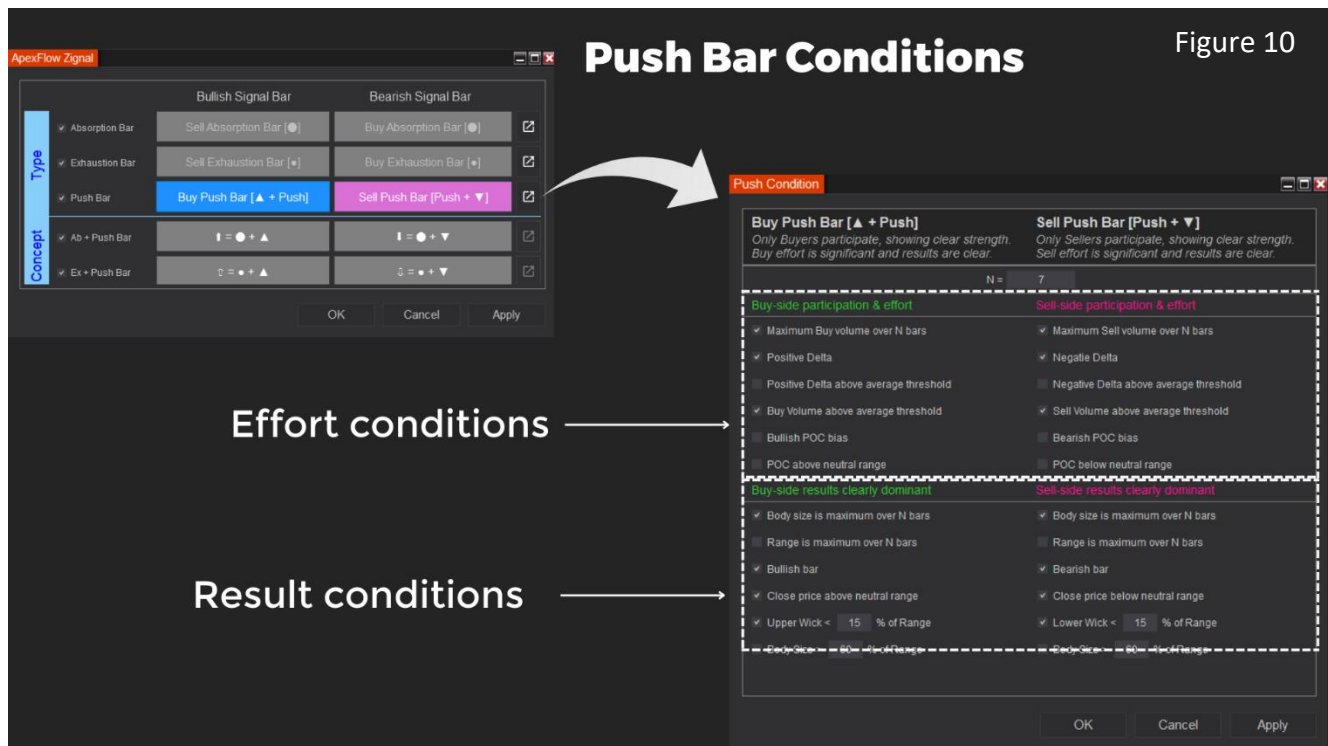
A Buy Exhaustion Bar is a **bearish** signal.



Decreasing Buy-side momentum is reflected through conditions such as:

- A sequence of N consecutive bullish bars
- A sequence of N bars with progressively smaller bullish bodies
- A sequence of N bars with progressively smaller full ranges
- Total volume decreasing within the recent N-bar context
- Buy volume decreasing within the recent N-bar context
- Positive delta exceeding its average threshold
- Buy volume exceeding its average threshold

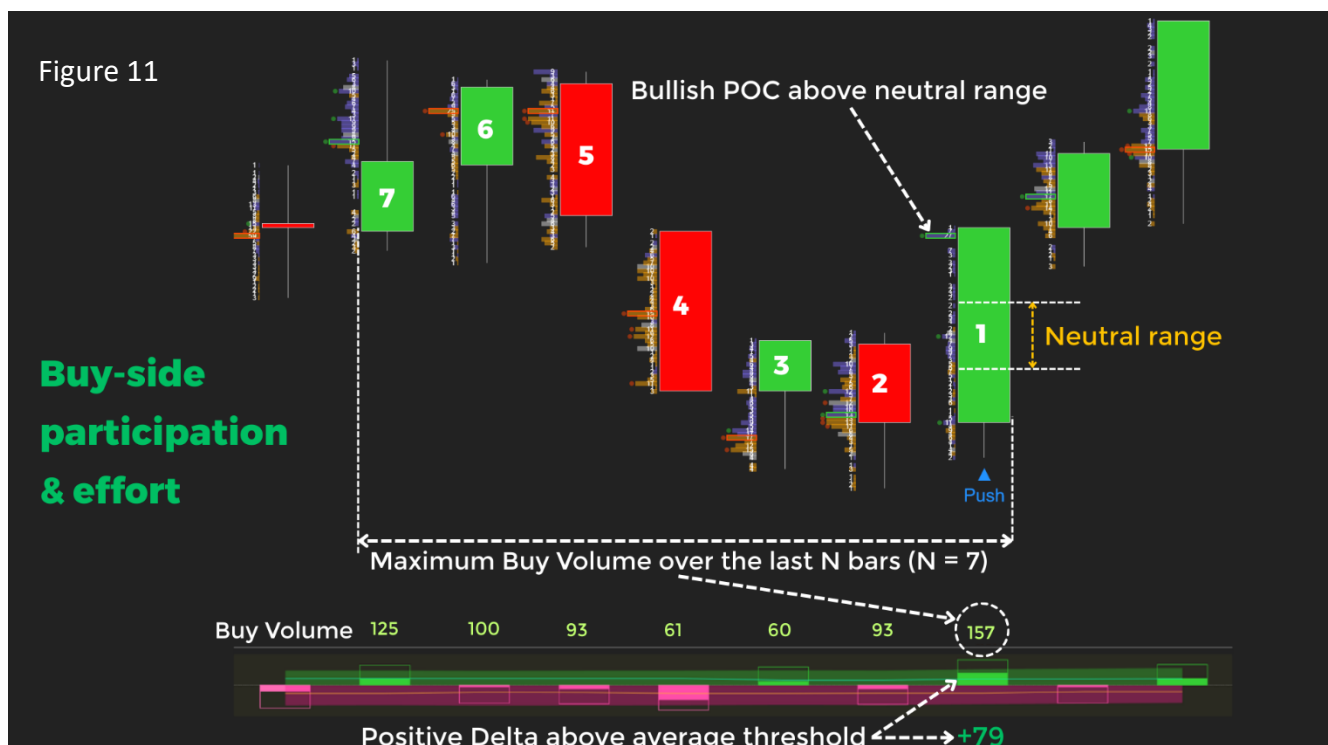
2.1.3. Push Bar



A **Push Bar** is a bar that reflects a strong alignment between **Effort & Result**.

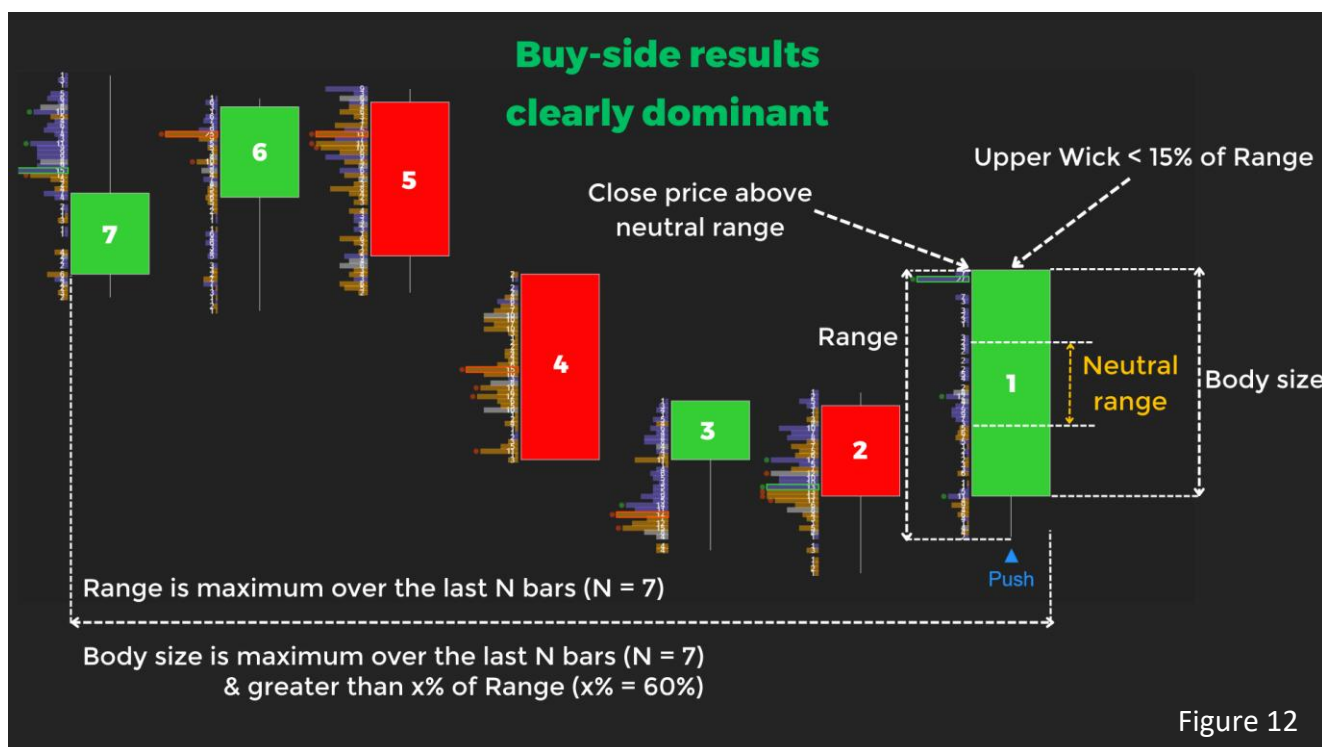
In a Push Bar, one side of the market (Buy or Sell) demonstrates **dominant** trading effort, and price responds clearly in the **same direction**, with no meaningful signs of absorption or rejection from the opposing side.

- **Buy Push Bar** appears when the Buy side is the active, controlling force in the order flow and price responds effectively with upward movement.



Buy-side participation & effort are reflected through conditions such as:

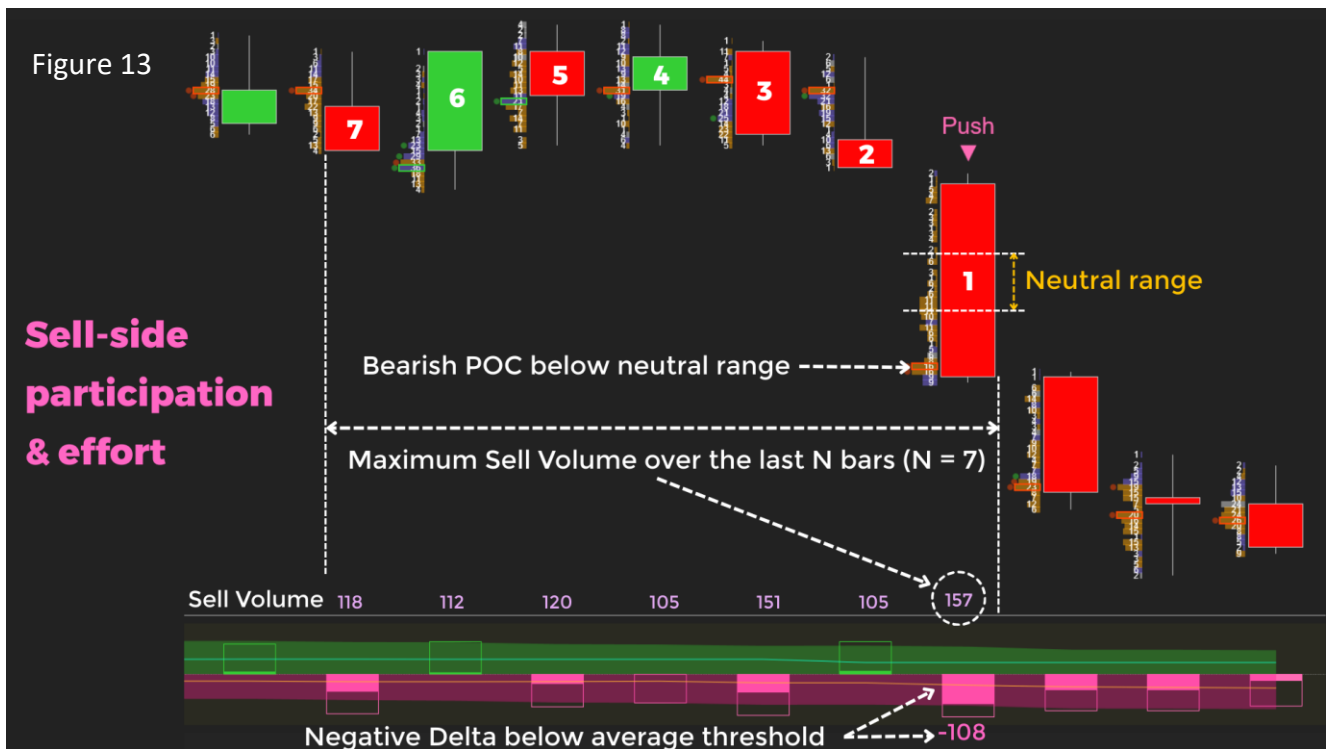
- Highest Buy volume within the recent N-bar context
- Positive delta
- Positive delta exceeding its average threshold
- Buy volume exceeding its average threshold
- Volume imbalance at the POC favoring the Buy side
- POC positioned above the Neutral Range



Buy-side results are clearly dominant, reflected through the following conditions:

- The bar has the largest body within the recent N-bar context
- The bar has the largest full range within the recent N-bar context
- The bar is bullish
- The closing price is above the Neutral Range
- The upper wick occupies a small proportion of the bar's total range
- The body occupies a large proportion of the bar's total range

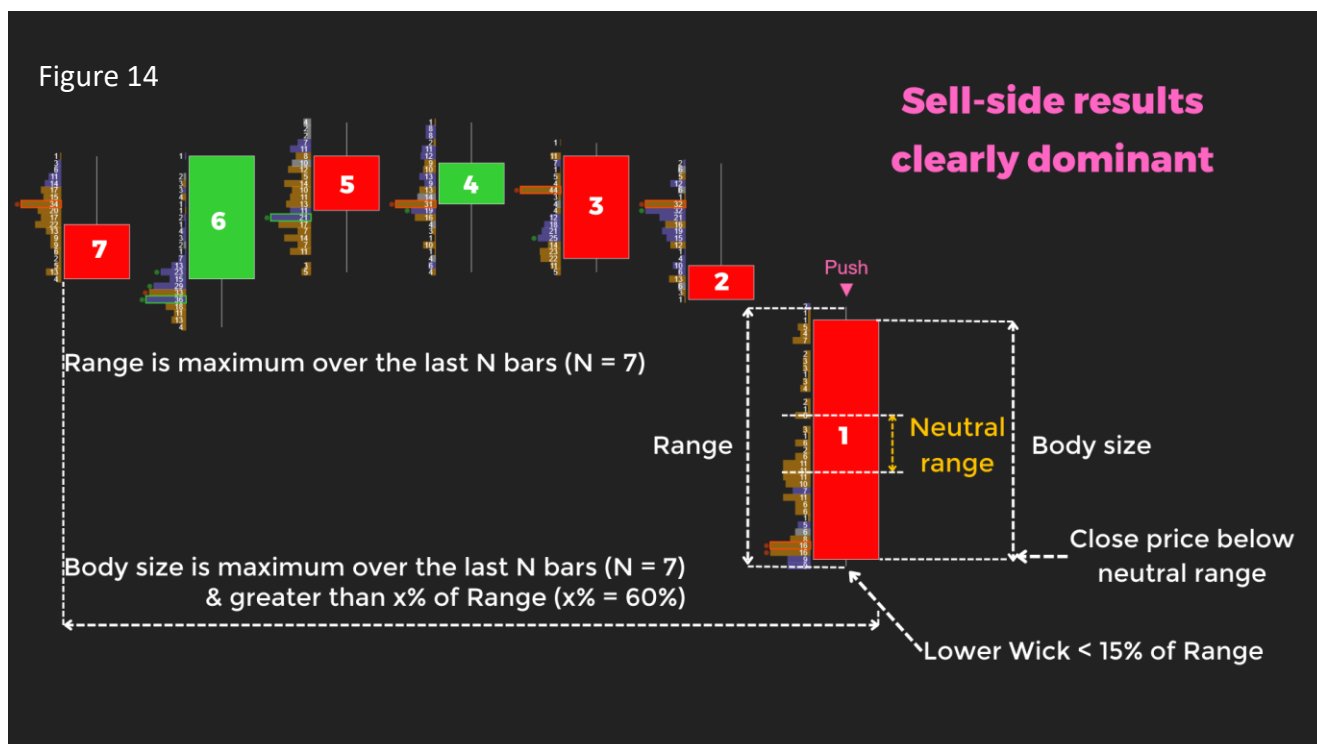
- **Sell Push Bar** appears when the Sell side is the active, controlling force in the order flow, and price responds effectively with downward movement.



Sell-side participation & effort are reflected through conditions such as:

- Highest Sell volume within the recent N-bar context
- Negative delta
- Negative delta exceeding its average threshold
- Sell volume exceeding its average threshold
- Volume imbalance at the POC favoring the Sell side
- POC positioned below the Neutral Range

Figure 14



Clearly dominant Sell-side results are reflected through the following conditions:

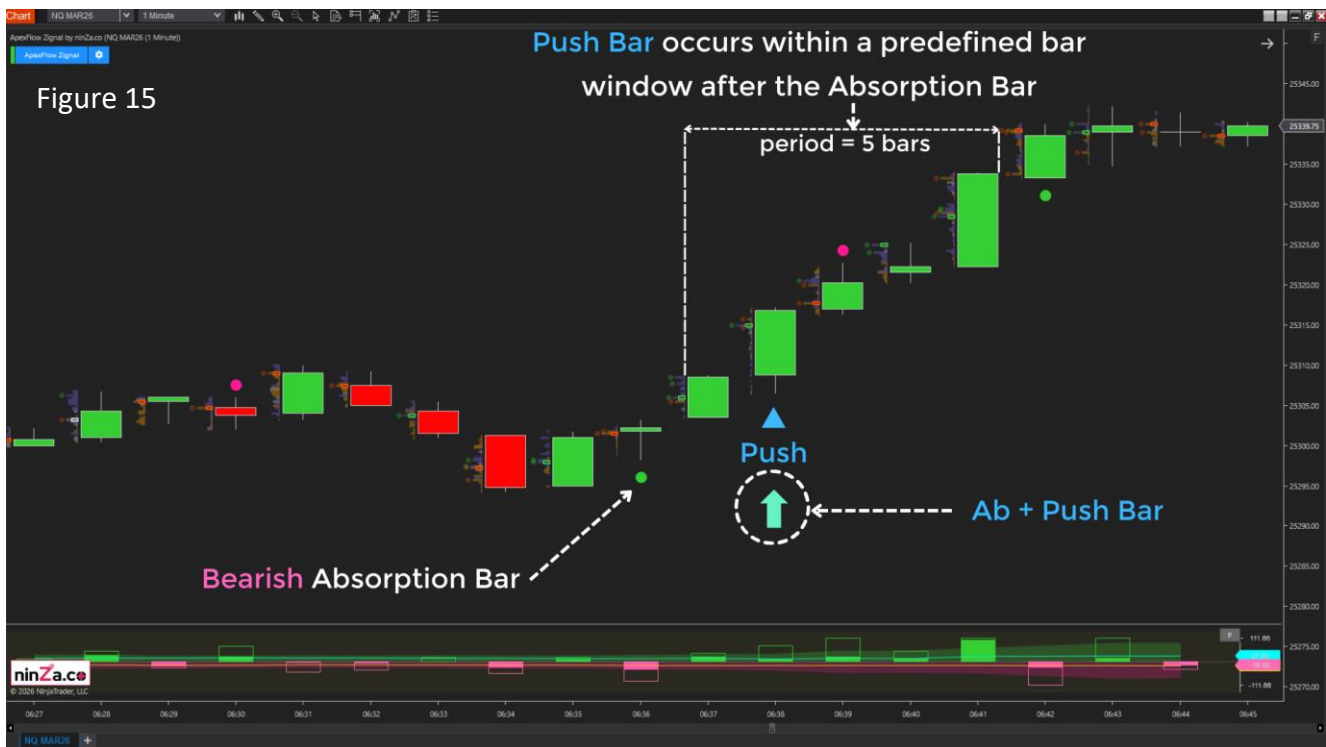
- The bar has the largest body within the recent N-bar context
- The bar has the largest full range within the recent N-bar context
- The bar is bearish
- The closing price is below the Neutral Range
- The lower wick occupies a small proportion of the bar's total range
- The body occupies a large proportion of the bar's total range

2.2. Signal Concepts

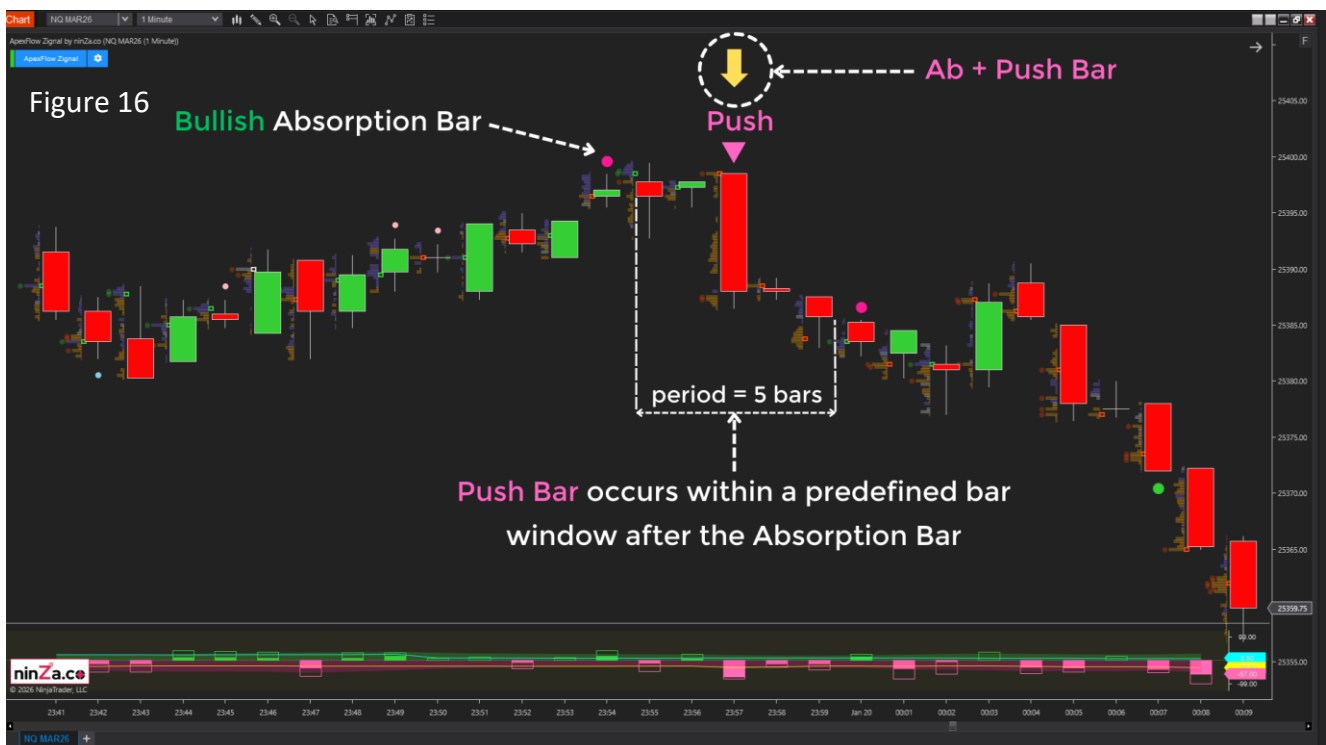
2.2.1. Ab + Push Bar

The **Ab + Push Bar** is a combined signal that represents a sequence of **order-flow absorption followed by a clear release of momentum**. In this pattern, an initial trading effort is absorbed, and the market subsequently produces a decisive push.

This signal confirms that effort was first absorbed and retained before being converted into a clear Result through a Push Bar.



- A **Buy Ab + Push Bar** forms in chronological order and consists of 2 consecutive stages:
 - A Sell Absorption Bar appears.
 - A Buy Push Bar appears within a defined time window (measured in number of bars) after the Sell Absorption Bar.

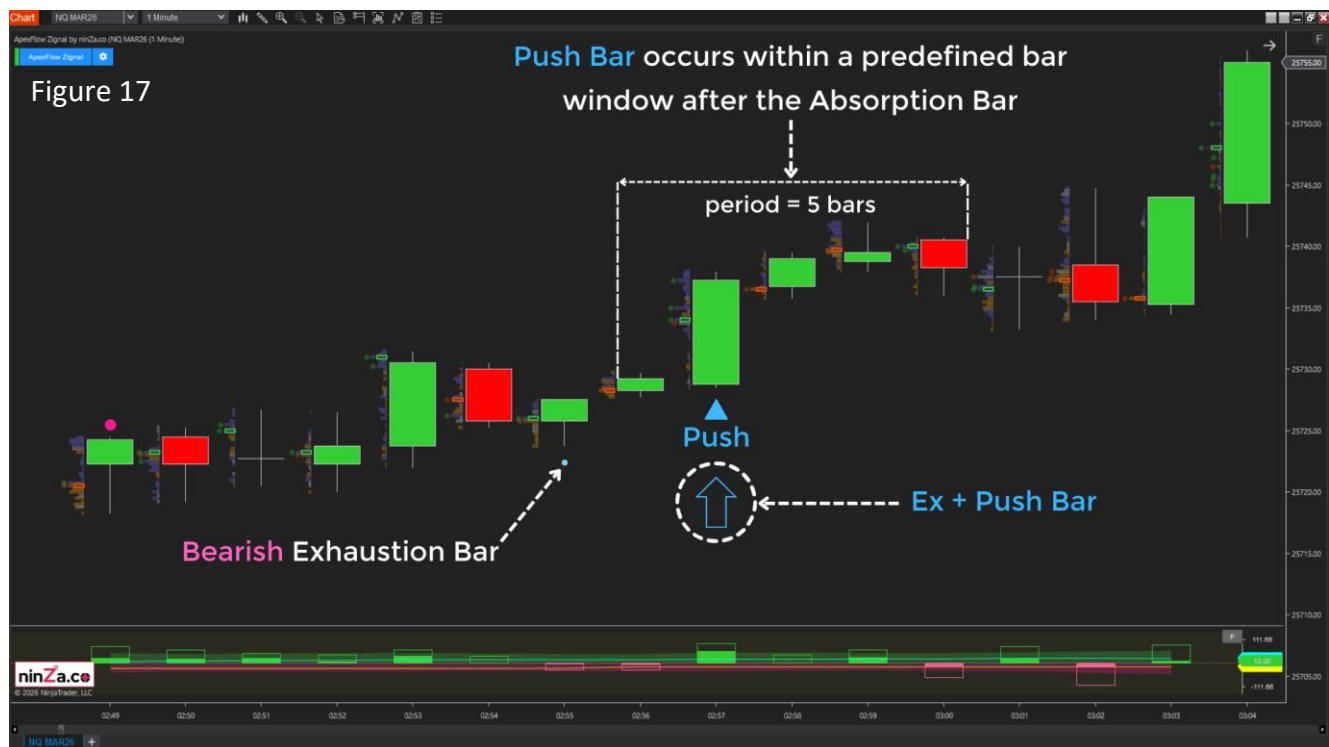


- A **Sell Ab + Push Bar** forms in chronological order and consists of 2 consecutive stages:
 - A Buy Absorption Bar appears.
 - A Sell Push Bar appears within a defined time window (measured in number of bars) after the Buy Absorption Bar.

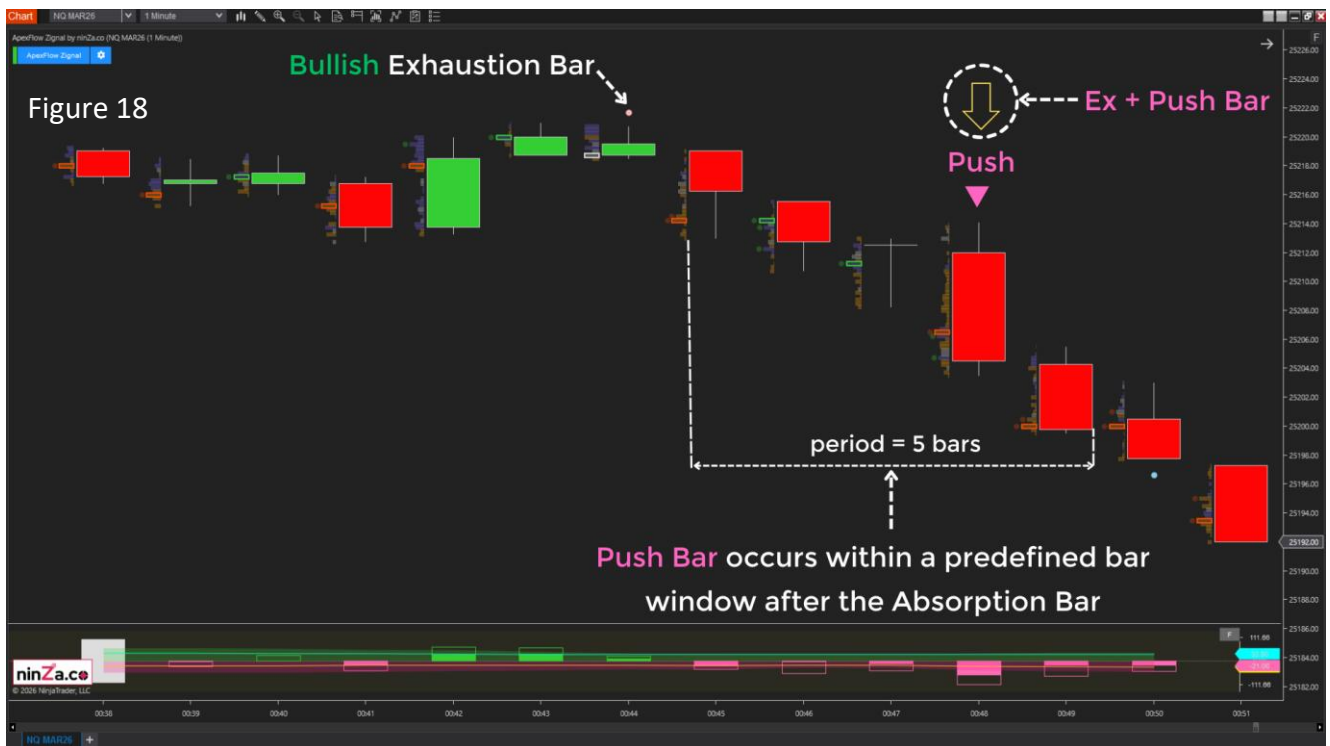
2.2.2. Ex + Push Bar

The **Ex + Push Bar** is a combined signal that illustrates a **transition of order-flow control**. In this pattern, the previous dominant side loses momentum, and the opposing side subsequently produces a clear Push phase.

This signal confirms not only that the market has lost momentum in the previous direction, but also that a new Effort has emerged strong enough to produce a clear Result in the opposite direction.



- A **Buy Ex + Push Bar** forms in chronological sequence and consists of 2 consecutive stages:
 - A Sell Exhaustion Bar appears.
 - A Buy Push Bar appears within a defined time window (measured in number of bars) immediately after the Sell Exhaustion Bar.

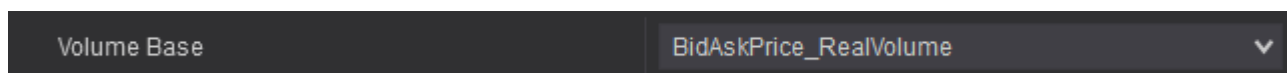


- A **Sell Ex + Push Bar** forms in chronological sequence and consists of 2 consecutive stages:
 - A Buy Exhaustion Bar appears.
 - A Sell Push Bar appears within a defined time window (measured in number of bars) after the Buy Exhaustion Bar.

3. Parameter

Parameters	
Volume Base	BidAskPrice_RealVolume
Avg Volume Period	10
Avg Delta Period	10
Neutral Range (%)	30
N: Absorption	3
N: Exhaustion	3
N: Push	7
Push: Min Body (Ticks)	10
Push Bullish: Max Upper Wick (%)	15
Push Bullish: Min Body (%)	60
Push Bearish: Max Lower Wick (%)	15
Push Bearish: Min Body (%)	60
Signal: Ab + Push Period (Bars)	5
Signal: Ex + Push Period (Bars)	5

3.1. Volume Base



- **BidAskPrice_RealVolume**

This mode should be used for instruments with real Volumes (**futures & stocks**).

Buy Volume & Sell Volume are defined as follows:

- **Buy Volume:** If the price of a tick is greater than or equal to the ask price, the real Volume of the tick is categorized as Buy Volume and added to the total Buy Volume of the bar that the tick belongs to.
- **Sell Volume:** If the price of a tick is less than or equal to the bid price, the real Volume of the tick is categorized as Sell Volume and added to the total Sell Volume of the bar that the tick belongs to.

In this mode, if Tick Replay is enabled, the indicator functions on both **historical & real-time data**. Please visit https://ninjatrader.com/support/helpGuides/nt8/en-us/?tick_replay.htm for instructions on how to enable Tick Replay.

If Tick Replay is disabled, the indicator functions only on **real-time data** and there are no calculations on historical data. Any actions that lead to chart reloading will erase all calculations.

- **UpDownTick_RealVolume**

This mode should be used for instruments with real Volumes (**futures & stocks**).

Buy Volume & Sell Volume are defined as follows:

- **Buy Volume:** If a tick is up or remains up, the real Volume of the tick is categorized as Buy Volume and added to the total Buy Volume of the bar that the tick belongs to.
- **Sell Volume:** If a tick is down or remains down, the real Volume of the tick is categorized as Sell Volume and added to the total Sell Volume of the bar that the tick belongs to.

In this mode, the indicator functions on both **historical & real-time data**. However, please be advised that historical and real-time calculations of the same bar may not be 100% identical, due to technical limitations.

- **UpDownTick_UnitVolume**

This mode is the one & only choice for instruments without real Volumes (**forex, CFDs, indices**).

Buy Volume & Sell Volume are defined as follows:

- **Buy Volume:** If a tick is up or remains up, a unit Volume (1) is credited as Buy Volume for the tick and added to the total Buy Volume of the bar that the tick belongs to.
- **Sell Volume:** If a tick is down or remains down, a unit Volume (1) is credited as Sell Volume for the tick and added to the total Sell Volume of the bar that the tick belongs to.

In this mode, the indicator functions on both **historical & real-time data**. However, please be advised that historical and real-time calculations of the same bar may not be 100% identical, due to technical limitations.

3.2. Avg Volume Period

"**Avg Volume Period**" defines the lookback period (number of bars) used to calculate the average volume for the Volume Average Area.

The Volume Average Area is calculated separately for each side of the market.

- **Volume Average Area: Buy:** The average threshold of Buy Volume.
- **Volume Average Area: Sell:** The average threshold of Sell Volume.

Increasing this value improves signal reliability but reduces the number of signals.

Decreasing this value increases the frequency of signals but may introduce more noise.

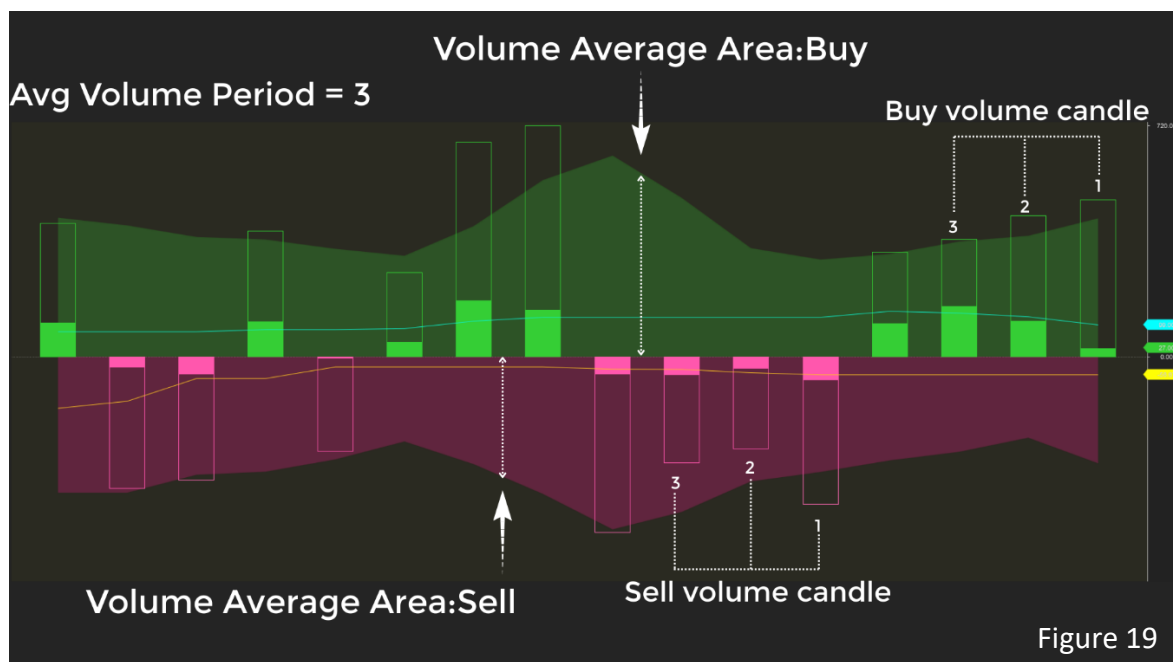


Figure 19

Example (Figure 19):

If **Avg Volume Period = 3**:

- **Volume Average Area: Buy** is calculated based on the Buy Volume of the most recent 3 bars.
- **Volume Average Area: Sell** is calculated based on the Sell Volume of the most recent 3 bars.

3.3. Avg Delta Period

"**Avg Delta Period**" defines the lookback period (number of bars) used to calculate the average Delta for the Delta Chart display.

The Delta average is calculated separately for the sides of the market.

- **Delta Threshold Positive:** The average threshold for positive Delta
- **Delta Threshold Negative:** The average threshold for negative Delta

Increasing this value improves signal reliability but reduces the number of signals.

Decreasing this value increases the frequency of signals but may introduce more noise.

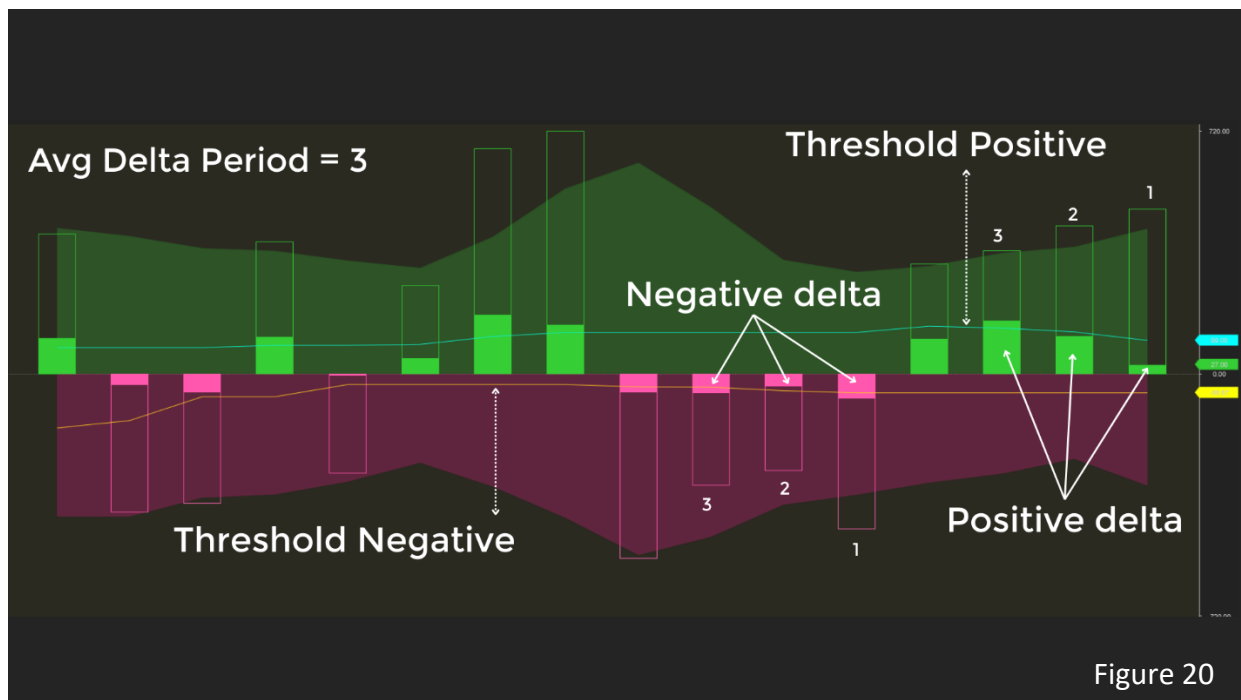


Figure 20

Example (Figure 20):

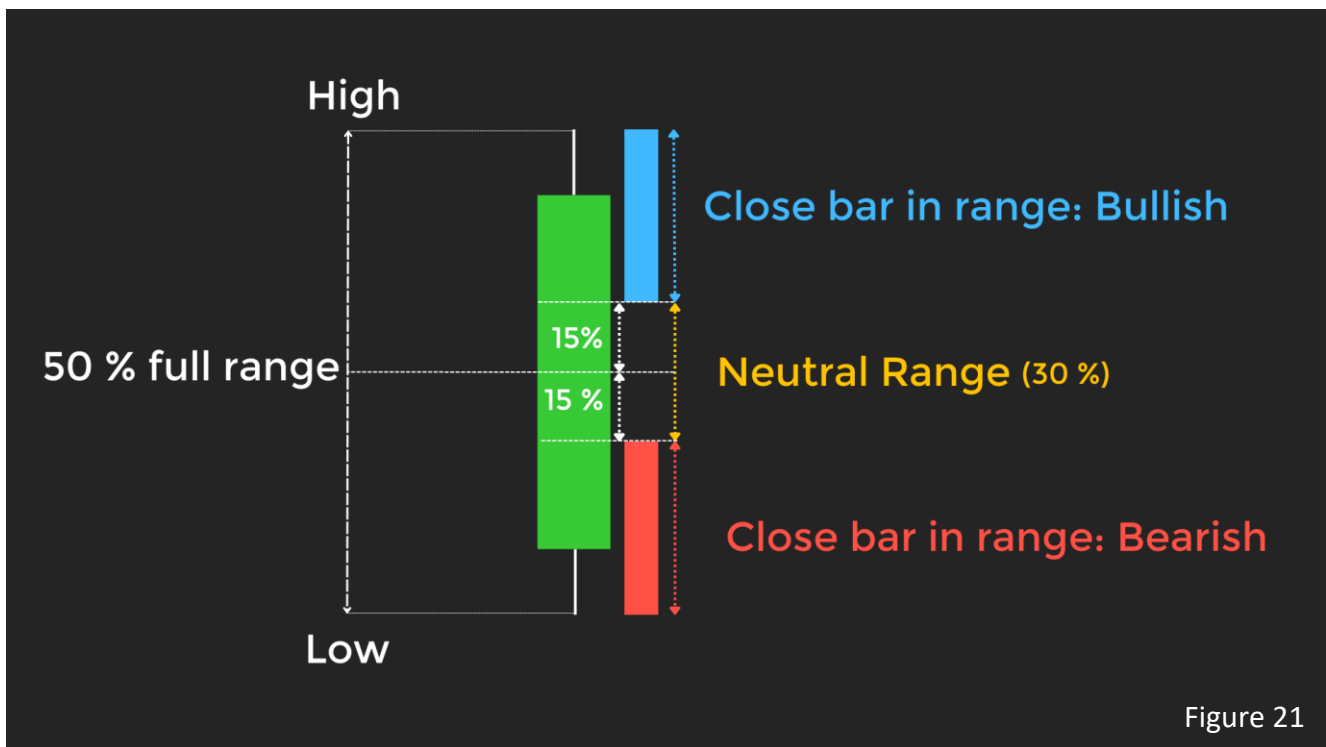
If Avg Delta Period = 3:

- **Delta Threshold Positive** is calculated based on the positive Delta of the most recent 3 bars.
- **Delta Threshold Negative** is calculated based on the negative Delta of the most recent 3 bars.

3.4. Neutral Range(%)

"**Neutral Range (%)**" defines the percentage threshold used to determine the neutral zone of a bar.

The Neutral Range is used in the algorithm to identify whether the Buy side or Sell side is dominant within a bar, based on the position of the closing price relative to the bar's full range.



Increasing this value improves signal reliability but reduces the number of signals.

Decreasing this value increases the frequency of signals but may introduce more noise.

Example (Figure 21):

If **Neutral Range (%) = 30**, the indicator calculates a neutral zone centered around the **50%** level of the bar's total range, with the upper and lower boundaries extending **15%** above and below that midpoint.

3.5. Signal Period (N)

Sell Absorption Bar [●]	Buy Absorption Bar [●]
Strong Sell effort, but the outcome favors the Buy side and is insignificant.	Strong Buy effort, but the outcome favors the Sell side and is insignificant.
N = 3	

- **"N: Absorption"** defines the lookback period (number of bars) used in calculating the Absorption Bar.

Example:

If **N = 3**, then any Absorption Bar conditions that use the N parameter will compare the current bar with the two preceding bars. This includes:

- The current bar
- The 2 bars immediately before it

Sell Exhaustion Bar [●] <i>Sell-side effort persists but momentum fades and exhausts.</i>	Buy Exhaustion Bar [●] <i>Buy-side effort persists but momentum fades and exhausts.</i>
N = 3	

- **"N: Exhaustion"** defines the lookback period (number of bars) used in calculating the Exhaustion Bar.

Example:

If **N = 3**, then any Exhaustion Bar conditions that use the N parameter will compare the current bar with the 2 preceding bars. This includes:

- The current bar
- The 2 bars immediately before it

Buy Push Bar [▲ + Push] <i>Only Buyers participate, showing clear strength. Buy effort is significant and results are clear.</i>	Sell Push Bar [Push + ▼] <i>Only Sellers participate, showing clear strength. Sell effort is significant and results are clear.</i>
N = 7	

- **"N: Push"** defines the lookback period (number of bars) used in calculating the Push Bar.

Example:

If **N = 7**, then any Push Bar conditions that use the N parameter will compare the current bar with the 6 preceding bars. This includes:

- The current bar
- The 6 bars immediately before it

Increasing the value of **N** improves signal reliability but reduces the number of signals.

Decreasing the value of **N** increases the number of signals but may introduce more noise.

3.6. Push Bar

3.6.1. Push: Min Body (Ticks)

Push: Min Body (Ticks)	10
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"Push: Min Body (Ticks)" defines the minimum candle body size (measured in ticks) required for a bar to qualify as a Push Bar.

Increasing this value improves signal reliability but reduces the number of signals.

Decreasing this value increases the frequency of signals but may introduce more noise.

Example:

If **Push: Min Body (Ticks) = 10**, it means that only bars with a candle body ≥ 10 ticks will qualify for Push Bar conditions.

3.6.2. Push: Min Body (%)

"Push: Min Body (%)" defines the minimum candle-body size (as a percentage of the candle's full range) required for a bar to qualify as a Push Bar.

Increasing these values makes the signals more reliable but reduces the number of signals.

Decreasing these values produces more signals, but they can become noisier.

Push Bullish: Min Body (%)	60
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- **"Push Bullish: Min Body (%)"** specifies the minimum body-percentage required for a Buy Push Bar, ensuring that the bullish push is strong and clearly dominant.

Push Bearish: Min Body (%)	60
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- **"Push Bearish: Min Body (%)"** specifies the minimum body-percentage required for a Sell Push Bar, ensuring that the bearish push is strong and clearly dominant.

3.6.3. Push Bullish: Max Upper Wick

Push Bullish: Max Upper Wick (%)	15
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"Push Bullish: Max Upper Wick (%)" defines the maximum allowable upper-wick percentage (relative to the candle's full range) for a Buy Push Bar to be considered valid.

Increasing this value results in more frequent signals but with higher noise.

Decreasing this value improves signal quality but reduces frequency.

Example:

If **Push Bullish: Max Upper Wick (%) ≤ 15** , it means only Buy Push Bars with an upper wick $\leq 15\%$ of the candle's total range are considered valid.

3.6.4. Push Bearish: Max Lower Wick

Push Bearish: Max Lower Wick (%)	15
----------------------------------	----

"Push Bearish: Max Lower Wick (%)" defines the maximum allowable lower-wick percentage (relative to the candle's full range) for a Sell Push Bar to be considered valid.

Increasing this value results in more frequent signals but with higher noise.

Decreasing this value improves signal quality but reduces frequency.

Example:

If **Push Bearish: Max Lower Wick (%) ≤ 15** , it means only Sell Push Bars with a lower wick $\leq 15\%$ of the candle's total range are considered valid.

3.6.5. Signal: Ex + Push Period

Signal: Ab + Push Period (Bars)	5
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"Signal: Ex + Push Period (Bars)" defines the maximum number of bars allowed between an Exhaustion Bar and a subsequent Push Bar for an Ex + Push signal to be considered valid.

Increasing this value produces more signals but introduces additional noise.

Decreasing this value improves reliability but reduces signal frequency.

Example:

If **Signal: Ex + Push Period (Bars) = 5**, it means that after an Exhaustion Bar forms, a Push Bar must appear within the next 5 bars for the Ex + Push signal to be considered valid.

3.6.6. Signal: Ab + Push Period

Signal: Ex + Push Period (Bars)	5
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"Signal: Ab + Push Period (Bars)" defines the maximum number of bars allowed between an Absorption Bar and a subsequent Push Bar for an Ab + Push signal to be considered valid.

Increasing this value produces more signals but adds noise.

Decreasing this value improves reliability but reduces signal frequency.

Example:

If **Signal: Ab + Push Period (Bars) = 5**, it means that after an Absorption Bar forms, a Push Bar must appear within the next 5 bars for the Ab + Push signal to be considered valid.